

Activity

Case Analysis – Avoiding Misleading Proportions

Question 1: Truncated Y-Axis in Sales Growth

A retail company presents the following monthly sales (in ₹ lakhs):

Month	Sales
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January	95
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February	97
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March	99
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April	101
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May	103
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June	105
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The marketing team creates a bar chart with the **Y-axis starting at 90 instead of 0**. The chart visually suggests that June sales are almost twice those of January.

Tasks:

1. Explain why the chart is misleading.

The bar chart is misleading because the Y-axis starts at 90 instead of 0. This makes the difference between January sales (₹95 lakhs) and June sales (₹105 lakhs) appear much larger than it actually is. In reality, the increase is only ₹10 lakhs, not almost double.

2. Identify the visualization principle that has been violated.

The violated principle is using an appropriate axis scale. Bar charts should normally start the Y-axis at zero so that the bar lengths accurately represent the data values.

3. Recommend an improved visualization method.

A bar chart with the Y-axis starting at zero is the most appropriate choice. A line chart can also be used because it clearly shows the gradual increase in monthly sales without exaggerating the differences.

4. Describe how this misleading chart could influence business decisions.

This chart may lead managers to believe that sales have increased dramatically. As a result, they may make poor business decisions, such as increasing production, inventory, or marketing expenses unnecessarily.

Question 2: Pie Chart with Incorrect Proportions

A university reports student enrollment in four departments.

Department	Students
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Computer Science	420
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Electronics	280
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Mechanical	200
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Civil	100
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The designer accidentally creates a pie chart where the slices occupy approximately **50%, 30%, 15%, and 15%** instead of the correct proportions.

Tasks:

1. Calculate the correct percentage for each department.

Department	Students	Percentage
Computer Science	420	42%
Electronics	280	28%
Mechanical	200	20%
Civil	100	10%

2. Explain why the displayed chart is misleading.

The pie chart is misleading because the slice sizes do not match the actual percentages. The chart shows approximately 50%, 30%, 15%, and 15%, while the correct proportions are 42%, 28%, 20%, and 10%. This gives viewers an incorrect understanding of the student distribution among departments.

3. Suggest an alternative chart that would improve comparison accuracy.

A bar chart is a better choice because it allows easy and accurate comparison of student enrollment across departments. The equal-width bars make differences in values easier to understand than pie chart slices.

4. Discuss how such errors could affect institutional planning.

An incorrect chart may lead the university to misunderstand student enrollment. As a result, resources such as faculty members, classrooms, laboratories, and budgets may be allocated incorrectly, affecting academic planning and decision-making.

Question 3: Unequal Bar Widths in Revenue Comparison

A software company compares annual revenue.

Product Revenue (₹ Crores)

A	60
B	75
C	90
D	105

Instead of using equal-width bars, the designer creates bars with increasing widths, making Product D appear nearly **three times larger** than Product A.

Tasks:

1. Identify the graphical error.

The graphical error is the use of unequal bar widths in the bar chart. The bars should all have the same width, but the designer made the widths increase, making Product D appear much larger than it actually is.

2. Explain how unequal bar widths distort perception.

Unequal bar widths make viewers judge both the height and width of the bars instead of only the height. As a result, Product D appears nearly three times larger than Product A, even though its revenue is only ₹105 crores compared to ₹60 crores.

3. Redesign the visualization by specifying the correct design principles.

The visualization should be redesigned using:

- Equal-width bars for all products.
- A common baseline starting from zero.
- A consistent scale on the Y-axis.
- Clear labels and a descriptive chart title.
- Appropriate colors without exaggerating differences.

4. Discuss the possible consequences if investors rely on this chart.

If investors rely on this misleading chart, they may believe that Product D generates significantly more revenue than the other products. This could result in poor investment decisions, incorrect budget allocation, and an inaccurate evaluation of the company's performance.

Question 4: 3D Column Chart Distortion

An energy company reports renewable energy generation.

Source	Energy (GWh)
Solar	450
Wind	500
Hydro	550
Biomass	520

The analyst uses a **3D column chart** with perspective effects. Due to the viewing angle, the Hydro bar appears almost **40% larger** than Wind, although the actual difference is only **10%**.

Tasks:

1. Explain why 3D charts can create misleading proportions.

3D charts use perspective and depth effects, which can make some bars appear larger or smaller than their actual values. The viewing angle distorts the visual comparison, making it difficult to judge the true differences between data values.

2. Compare the perceived difference with the actual difference.

The chart makes the Hydro bar appear almost 40% larger than the Wind bar. However, the actual energy values are 550 GWh for Hydro and 500 GWh for Wind, which is only a 10% difference. This exaggerates the comparison and can mislead viewers.

3. Recommend a better visualization.

A 2D bar chart is a better choice because it provides an accurate comparison of energy generation without perspective distortion. All bars should have equal widths and start from a common baseline.

4. Discuss the importance of avoiding visual distortion in sustainability reporting.

Accurate visualizations are essential in sustainability reporting because they help stakeholders make informed decisions. Misleading charts can result in incorrect conclusions about renewable energy performance, affecting planning, investment, and policy decisions.

Question 5: Bubble Chart with Incorrect Scaling

A healthcare organization compares disease cases across five districts.

District Cases

A	150
B	250
C	350
D	450
E	550

The analyst scales the **bubble radius directly** to the number of cases rather than scaling the **bubble area**. As a result, District E appears more than **ten times larger** than District A.

Tasks:

1. Explain why scaling the radius instead of the area is misleading.

A bubble chart should represent data using the area of each bubble. If the radius is scaled directly according to the number of cases, the bubble area increases much more rapidly, making larger values appear greatly exaggerated. This causes District E to look much larger than its actual proportion compared to District A.

2. Describe the correct method for proportional bubble sizing.

The correct method is to scale the area of each bubble in proportion to the data value. The radius should be calculated from the area so that the visual size accurately represents the number of cases.

3. Suggest an alternative visualization suitable for this dataset.

A bar chart is a better visualization because it provides a clear and accurate comparison of disease cases across all districts without causing size distortion.

4. Discuss the potential impact of this visualization on public health resource allocation.

A misleading bubble chart may cause decision-makers to overestimate or underestimate the number of disease cases in different districts. This could lead to the incorrect allocation of medical staff, hospital beds, medicines, and healthcare funds, reducing the effectiveness of public health planning.